SMART TEMPERATURE & HUMIDITY MONITORING SYSTEM

Internship Domain: Embedded Systems with Arduino
Submitted by: SEELAM HARI VENKAT
Department: Electronics and Communication Engineering
College: SRKR Engineering College
Organization: VAIDSYS Technologies (VST)

1. Project Overview

This project is an embedded system developed using Arduino Uno and a DHT22 sensor to monitor environmental temperature and humidity. The measured values are displayed on a 16×2 I2C LCD. A buzzer alerts the user whenever the temperature exceeds 30°C, enabling continuous environmental monitoring.

2. Introduction

Embedded systems are dedicated computing systems designed for specific tasks. This project demonstrates how Arduino can interface with sensors and output devices to build a reliable monitoring system suitable for homes, laboratories, greenhouses and industries.

3. Problem Statement

Manual monitoring of environmental conditions is time-consuming and may lead to delayed responses. The project automates temperature and humidity monitoring and provides an instant warning when the temperature exceeds the preset limit.

4. Objectives

• Monitor temperature and humidity in real time.
• Display readings on an LCD.
• Generate an alert using a buzzer.
• Demonstrate practical embedded systems programming with Arduino.

5. Hardware & Software Requirements

Hardware: Arduino Uno, DHT22 Sensor, 16×2 I2C LCD, Active Buzzer, Breadboard, Jumper Wires, USB Cable.
Software: Arduino IDE, Embedded C++, DHT Library, Wire Library and LiquidCrystal_I2C Library.

6. Working Principle

The DHT22 sensor measures temperature and humidity. Arduino reads these values every two seconds, displays them on the LCD, and compares the temperature with a threshold of 30°C. If the threshold is crossed, the buzzer turns ON; otherwise it remains OFF.

7. Arduino Program & Explanation

The setup() function initializes the LCD, DHT22 sensor and buzzer. The loop() function continuously reads sensor values, updates the LCD display and controls the buzzer using a simple conditional statement based on the temperature threshold.

8. Project Implementation & Results

The prototype was assembled successfully. During testing, the LCD displayed live temperature and humidity values accurately. The buzzer activated whenever the measured temperature exceeded 30°C, confirming correct operation.

9. Applications

Smart homes, weather monitoring, laboratories, server rooms, greenhouses, warehouses and industrial safety monitoring.

10. Conclusion

The Smart Temperature & Humidity Monitoring System successfully demonstrates real-time environmental monitoring using Arduino. The project improved practical knowledge of sensor interfacing, embedded programming and hardware integration.

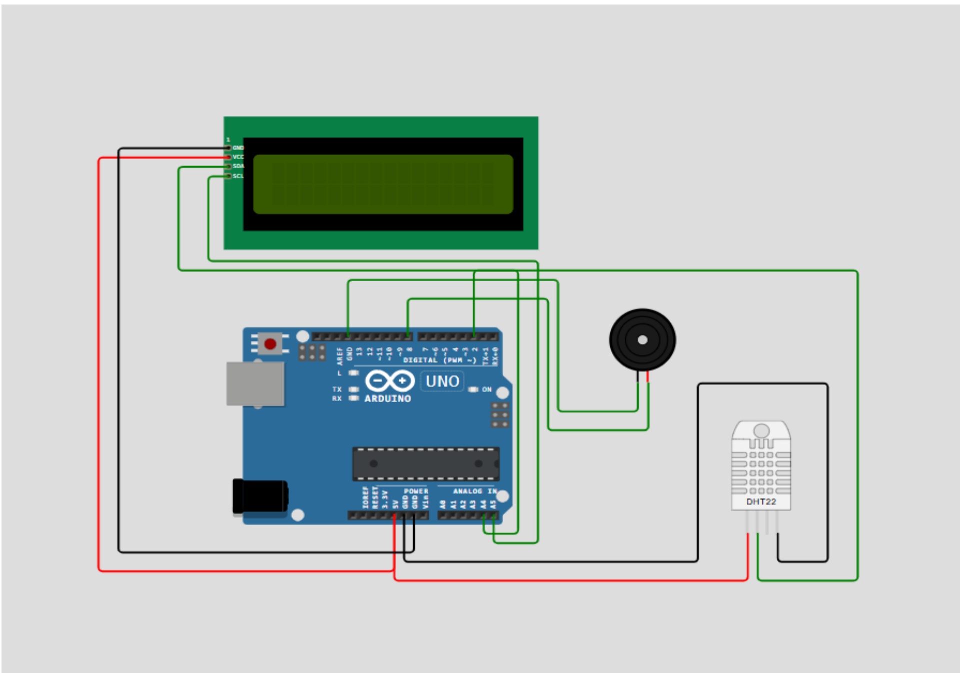

Figure 1: Circuit setup.

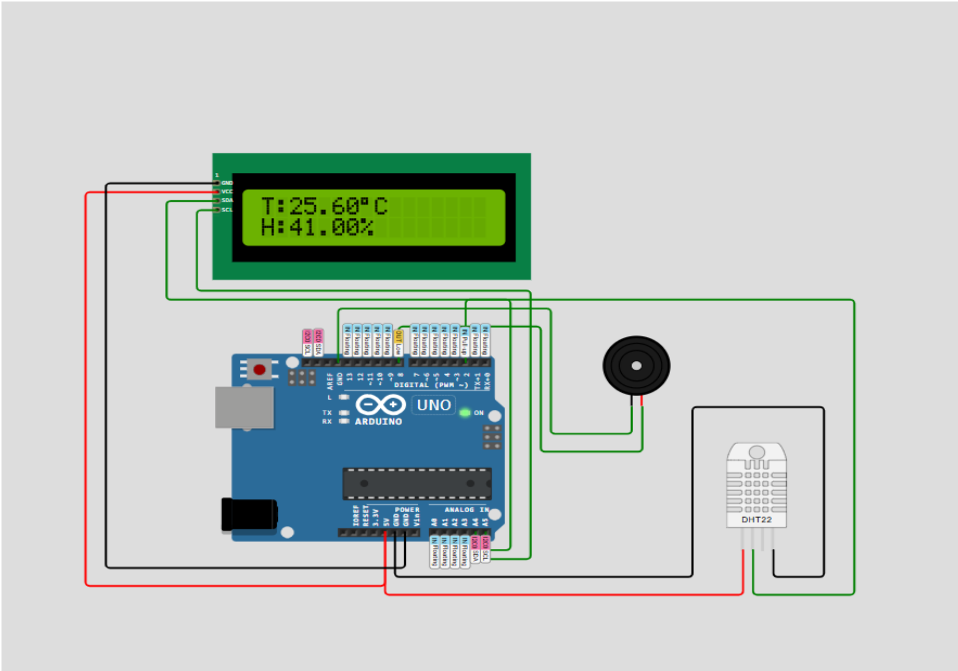

Figure 2: Normal operating condition.

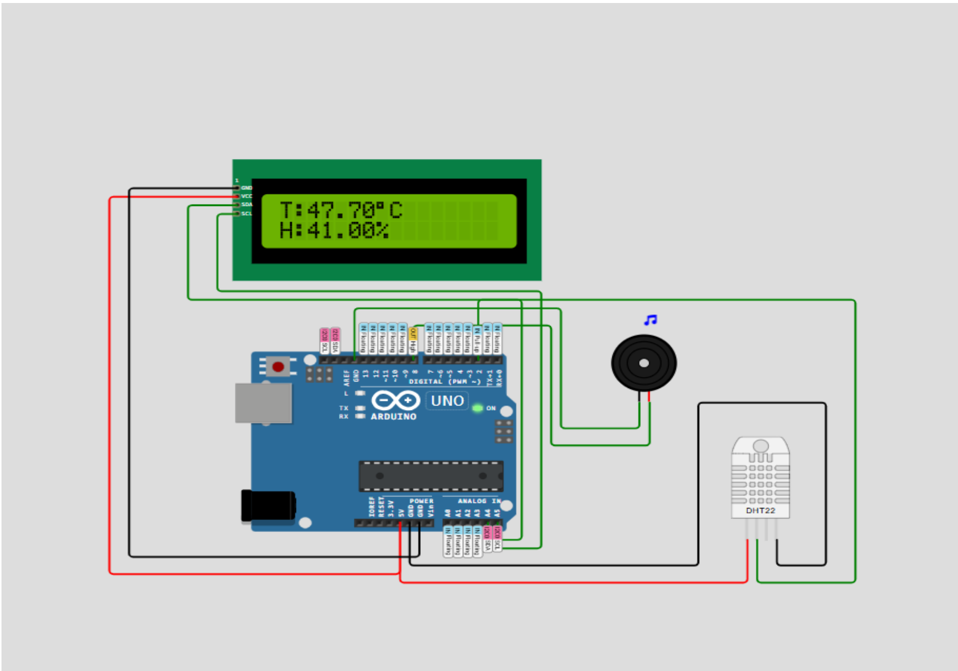

Figure 3: High temperature alert condition.

Appendix

test

Arduino Program

```
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define DHTPIN 2
#define DHTTYPE DHT22
#define BUZZER 8

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
pinMode(BUZZER, OUTPUT);
digitalWrite(BUZZER, LOW);

dht.begin();

lcd.init();
lcd.backlight();

lcd.setCursor(0, 0);
lcd.print("Temp & Humidity");
lcd.setCursor(0, 1);
lcd.print("Monitoring...");
delay(2000);
}

void loop() {
float temperature = dht.readTemperature();
float humidity = dht.readHumidity();

lcd.clear();

lcd.setCursor(0, 0);
lcd.print("T:");
lcd.print(temperature);
lcd.print((char)223);
lcd.print("C");

lcd.setCursor(0, 1);
lcd.print("H:");
lcd.print(humidity);
lcd.print("%");

if (temperature > 30) {
tone(BUZZER, 1000);
} else {
noTone(BUZZER);
}

delay(2000);
}
```